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Performance analysis and particle swarm optimization of molten salt-based nanofluids in parabolic trough concentrators

Amr Kaood [a], Mohamed Abubakr [b], Otabeh Al-Oran [c, d], Muhammed A. Hassan [e, *]

[a] Mechanical Engineering Department, Faculty of Engineering, Fayoum University, El-Fayoum, 63514, Egypt
[b] Mechanical Design and Production Engineering Department, Faculty of Engineering, Cairo University, Giza, 12613, Giza, Egypt
[c] Department of Energy Engineering, Faculty of Mechanical Engineering, Budapest University of Technology and Economics, Muegyetem rkp. 3, Budapest, 1111, Hungary
[d] Mechanical Engineering Department, The University of Jordan, Amman, 11942, Jordan
[e] Mechanical Power Engineering Department, Faculty of Engineering, Cairo University, Giza, 12613, Giza, Egypt

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ABSTRACT

Molten salts are typically used as energy storage media in concentrating solar power systems for their lower costs and environmental impact. This study aims to map and optimize the performance of parabolic trough concentrators (PTCs) working with molten salt-based nanofluids (MSNFs) as heat transfer media at high temperatures. The thermal, hydraulic, energetic, and exergetic performances were analyzed and optimized using a unique framework of Monte Carlo optical simulations, computational fluid dynamics, data-drive support vector regression, particle swarm optimization, and decision-making techniques. Three molten salts (Solar Salt, Hitec, and Hitec XL) and three nanoparticle types (Al_2O_3, CuO, and SiO_2) were investigated in a broad range of volumetric concentrations (0.0–4.0%), operating Reynolds numbers (4×10^3 to 40×10^3), and temperatures (535–805 K). The results showed a maximum energy efficiency of 69.1%, achieved when using SiO2-Hitec nanofluid (1.0%) at a Reynolds number of 40×10^4 and temperature of 535 K. The maximum achieved exergy efficiency was 70.48%, obtained using pure Hitec at a Reynolds number of 40×10^4 and temperature of 535 K. The maximum possible enhancements in energy and exergy efficiencies in the covered range are 17.0 and 42.0%, respectively. The optimal combination of energy and exergy efficiencies are ~73.1 and 69.0%, obtained using CuO-Hitec nanofluid at temperature, Reynolds number, and concentration of 535 K, 39912.98, and 0.019%, respectively. The optimum combination of percentage enhancements in energy and exergy efficiencies are 0.465 and 7.182%, respectively, which corresponds to CuO-Hitec nanofluid operating at 805 K, 32025.4, and 0.092%, respectively.

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1. Introduction

Parabolic trough collectors (PTCs) are promising line-focus energy converters for concentrating solar power systems that have been used to produce electricity when utilized and synchronized with a power block [1]. To enhance the productivity and reliability of PTCs, several passive and active performance enhancement methods have been proposed in the literature, as summarized in Refs. [2,3]. Such methods are generally classified as optical [4,5] or thermal [6,7] enhancement methods. The most notable thermal performance enhancement methods are finned tubes [7] and nanofluids [8].

Nanofluids have been used in PTCs as alternative fluids to enhance the thermal properties of synthetic oils through induced convection and collisions between the solid particles (≤100 nm) and the liquid phase of the mixture [9]. The use of nanofluids is still an experimental concept. Hence, the literature contains many studies on various nanoparticles that have been examined numerically [10–12] and experimentally [13] under different operating conditions, mostly resulting in a positive impact on the thermal performance and an adverse impact of the hydraulic PTC performance [14,15]. Gasemi and Rangbar [16] studied Al_2O_3/Therminol 66 nanofluid in the absorber tube of a PTC and reported a gradual increase of thermal efficiency by increasing the

* Corresponding author. #1 Cairo University Street, Giza, 12613, Giza, Egypt.
E-mail addresses: aki03@fayoum.edu.eg (A. Kaood), aloran@energia.bme.hu (O. Al-Oran), mhd.zidan17@cu.edu.eg, mhd.a.zidan@gmail.com (M.A. Hassan).

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nanoparticles' concentration. Mwesigye et al. [17] investigated the use of Cu/Therminol VP-1 in a PTC with different geometrical concentration ratios numerically. They demonstrated heat transfer enhancements up to 32% for a volumetric concentration of 6%. Benabdelrahmane et al. [18] examined a combined heat transfer enhancement approach by using internal fins and different nanoparticle types (metal, non-metal, and oxide) suspended in Dowtherm oil. Their model showed higher potential enhancements when both methods are applied, especially for metallic nanoparticles.

Molten salts are among the few candidate heat transfer media for the next generation of PTCs. They are particularly interesting due to their lower costs, higher specific heat capacities, higher stability at elevated operating temperatures (up to 600 °C), and lower adverse environmental impact [19]. Chang et al. [20] developed a finite volume model to examine the use of eccentric pipes for boosting the performance of PTCs working with molten salts. They achieved an improved performance of 64% and also an enhanced temperature distribution. In Ref. [21], Cheng et al. reduced the absorber tube temperature, improved the temperature distribution, and enhanced the PTC's performance by ~84% when operating with molten salts and fitted with rod inserts.

The promising results of oil-based nanofluids encouraged researchers to examine molten salt-based nanofluid (MSNF) [22]. Ho and Pan [23] investigated the impact of Al_2O_3 nanoparticle concentration in Hitec salt on a tube heat exchanger's performance. They showed an enhancement of 11.6% in the average Nusselt number for a concentration of 0.25 wt%. Ying et al. [24] used a computational model to study the effects of nanoparticle concentration, inlet velocity, and heat flux profile on the thermo-hydraulic performance of Al_2O_3/Hitec nanofluid in a generic linear solar receiver. The largest increases in Nusselt number (6.9%) and pressure drop were noticed when the volume fraction is 0.063%. Chen et al. carried out a similar experimental study [25] but using KNO_3–$Ca(NO_3)_2$ salt with 1.0 wt% of SiO_2. They reported enhancements of 16.3 and 39.9% in the Nusselt number and the corresponding heat transfer coefficient, respectively. Hu et al. [26] computationally studied SiO_2-Solar Salt in a tube. They reported a best-performing nanoparticle concentration of 1.0 wt%, for which Nusselt number and heat transfer coefficient were enhanced by 7.29 and 8.58%, respectively.

So far, most of the studies on MSNFs are concerned with preparing stable MSNFs and characterizing their thermophysical properties [27–29]. Studies focused on the use of MSNFs in the context of PTC systems are scarce. Bellos et al. [30] developed a computational model to analyze two CuO-based nanofluids (volume fraction of 6.0%) in LS-2 PTC. The base fluids were Syltherm 800 oil and Solar Salt, for which the Nusselt number was improved by 40.0 and 13.0%. Nevertheless, the corresponding thermal efficiencies were marginally enhanced by 0.76 and 0.26%. Lastly, Abed et al. [31] numerically examined and compared different combinations of nanoparticles (Al_2O_3, CuO, Fe_2O_3, TiO_2, CeO_2, and SiO_2) and base fluids (molten salt, Therminol VP-1, and water) in PTCs at different operating temperatures. They found that silicon dioxide (SiO_2) is the most effective nanofluid, regardless of the base fluid type.

Based on this literature survey, it can be seen that the performance of PTCs operating with MSNFs is rarely investigated in the open literature [32], and the few relevant studies examined only a few operating conditions. Hence, more studies are required to fully map and optimize the system performance. A possible technique to approach this problem is to use high-fidelity Monte Carlo ray tracing (MCRT)-finite volume (FVM) models to generate just a sufficient number of precise estimates of investigated parameters, and then to develop an accurate black-box model to act as a fineness function in the multi-objective optimization problem. Only one recent study by the authors (Abubakr et al. [6]) was carried out to simultaneously optimize the energy and exergy efficiencies of PTCs operating with oil-based nanofluids for different particle concentrations and operating conditions. They used a multi-layered feedforward backpropagation artificial neural network, alongside with NSGA II genetic algorithm, for optimizing the performance. They reported optimal energy and exergy efficiencies of 69.9 and 32.1% using CuO/Dowtherm nanofluid.

This study proposes a similar approach but for MSNFs, where the relevant literature is inadequately limited. Moreover, the least-squares support vector machine (SVM) algorithm is used for replicating the computational fluid dynamics (CFD) simulations and also for mapping the PTC performance. This algorithm is selected since it is inherently more stable than neural network algorithms [33]. Based on its concept of structural minimization, it basically solves convex problems with a single minimum. Besides, the multi-objective particle swarm optimization (MOPSO) algorithm is used for simultaneously optimizing the absolute and relative overall energetic and exergetic performances since it converges faster than NSGA-II [34]. This facilitates the process of tuning multiple controlled parameters and minimizing different response ones. To the best of the authors' knowledge, this is the first study in this context with such methodologies and objectives.

2. Physical system

2.1. Parabolic trough collector

A PTC consists of high-reflectively parabola-shaped mirrors and a linear heat collecting element (HCE), comprising a receiver tube, incubated in a glass envelope. The mirrors concentrate the direct irradiance component on the HCE. The HCE is responsible for absorbing the major part of concentrated sun rays and transferring the thermal power to the heat transfer fluid (HTF) flowing inside the receiver tube. To decrease the HCE heat losses, the tube is coated with a high absorptivity coating layer, and the space between the tube and the glass cover is evacuated ($\leq$0.001 mbar). Table 1 provides the main specifications of the physical PTC adopted in this study, namely Euro-trough PTC with PTR-70 HCE.

2.2. Nanofluids

In this study, three molten salts, namely: Solar Salt, Hitec, and Hitec XL, are considered to be the base fluid since they are widely used in thermal energy storage systems of concentrating power plants, as well as in the central receivers of the solar dish and solar tower technologies, for operating at high temperatures without phase-change or stability risks. Solar salt is a binary salt mixture

Table 1
Technical specifications of Euro-trough PTC and PTR70 HCE [35,36].

Component	Specification	Value
Mirrors	Width	5.76 m
	Focal length	1.71 m
	Reflectivity	0.94
Receiver Tube	Material	Steel
	Outer diameter	70 mm
	Inner diameter	65 mm
	Absorptivity	0.96
	Emissivity	0.095
Glass Cover	Material	Borosilicate glass
	Outer diameter	12.0 mm
	Inner diameter	11.5 mm
	Transmissivity	0.97

composed of sodium and potassium nitrates: $NaNO_3$–KNO_3 (60:40 wt%). Hitec is a ternary salt mixture of sodium and potassium alkali-nitrates, as well as sodium alkali-nitrites: $NaNO_3$–KNO_3–$NaNO_2$ (7:53:40 wt%). Hitec XL is also a ternary salt mixture but composed of sodium, potassium, and calcium nitrates: $NaNO_3$–KNO_3–$Ca(NO_3)_2$ (5:47:48 wt%). The three salts have melting points of 220, 142, and 120 °C, and stability limits of 600, 535, and 500 °C, respectively [37]. The following equations are used to describe the specific heat capacity (C_p in J/kg.K), mass density (ρ in kg/m^3), thermal conductivity (λ in W/m.K), and dynamic viscosity (μ in Pa.s) of the three base fluids (subscript: _bf) as functions of the operating temperature (T in K) [38].

Solar salt

$$C_{p,bf} = 1443 + 0.172(T - 273.15) \quad (1)$$

$$\rho_{bf} = 2090 - 0.636(T - 273.15) \quad (2)$$

$$\lambda_{bf} = 0.443 + 1.9 \times 10^{-4}(T - 273.15) \quad (3)$$

$$\mu_{bf} = 2.2714 \times 10^{-2} - 1.2 \times 10^{-4}\,(T - 273.15) + 2.281 \times 10^{-7}(T - 273.15)^2 - 1.474 \times 10^{-10}(T - 273.15)^3 \quad (4)$$

Hitec:

$$C_{p,bf} = 1.56 \times 10^{-3} \quad (5)$$

$$\rho_{bf} = -0.74(T - 273.15) + 2084 \quad (6)$$

$$\lambda_{bf} = 0.411 + 4.36 \times 10^{-4}(T - 273.15) - 1.54 \times 10^{-6}(T - 273.15)^2 \quad (7)$$

$$\mu_{bf} = 10^{2.7374}(T - 273.15)^{-2.104} \quad (8)$$

Hitec XL:

$$C_{p,bf} = -0.33T + 1634 \quad (9)$$

$$\rho_{bf} = 2240 - 0.827(T - 273.15) \quad (10)$$

$$\lambda_{bf} = 0.519 \quad (11)$$

$$\mu_{bf} = 10^{6.1374}(T - 273.15)^{-3.36406} \quad (12)$$

The correlations of the three molten salts are valid for the temperature ranges $533 < T < 873$ K; $415 < T < 808$ K, and $403\ K < T < 823$ K.

On the other hand, three nanoparticle types have been considered, namely: aluminum oxide (Al_2O_3), copper oxide (CuO), and silicon dioxide (SiO_2), which are commonly adopted in relevant literature with a good understanding of their behaviors. The thermal properties of those nanoparticles are provided in Table 2.

By studying three different base fluids and three different nanoparticles, there are nine nanofluids covered in this study. For each nanofluid, four volumetric concentrations of the nanoparticles are considered through a full factorial experimental design, namely: 0.5, 1.0, 2.0, and 4.0%.

Table 2
Thermal properties of adopted nanoparticles [8,39]. The subscript (_np) stands for nanoparticles.

Property	Unit	Al_2O_3	CuO	SiO_2
Specific heat capacity ($C_{p,np}$)	J/kg.K	773	551	765
Density (ρ_{np})	kg/m^3	3960	6000	3970
Thermal conductivity (λ_{np})	W/m.K	40	33	1.4

A homogenous single-phase approach was selected to tackle the problem. This approach discards the relative velocities and thermal imbalances between the solid (nanoparticles) and liquid (salt) phases of the nanofluid, resulting in drastically simpler governing equations that can be solved with lower computational costs. Ying et al. [40] evaluated this approach, alongside other multi-phase approaches, and found they are all reliable with maximum deviations of less than 20.0%. Similar conclusions were also reported by Mahian et al. [41], and the approach itself is widely used in relevant literature [24,26,30]. It is essential here to control the computational costs (as long as a good precision is maintained) due to the enormous number of performed simulations.

To determine the combined thermo-physical properties of the nine MSNFs, four models, widely used in the literature, have been carefully selected. Equation (13) adopts the mixing theory to express the nanofluid's density (ρ_{nf}) as a function of those of the base fluid and nanoparticles, as well as the volumetric nanoparticle concentrations (ϕ). The formula of nanofluid's specific heat capacity ($C_{p,nf}$) is provided in Equation (14). It is used here due to its accuracy for a wide range the volume concentrations. The thermal conductivity of the nanofluid (λ_{nf}) was obtained using the Maxwell model, as shown in Equation (15). Lastly, the Vajjha model was used to obtain the dynamic viscosity of the nanofluid (μ_{nf}), as presented in Equation (16). In this equation, A and B are nanoparticle constants, which respectively equal 1.092 and 5.954 for SiO_2, 0.983, and 12.959 for Al_2O_3, and 0.920 and 22.854 for CuO [6,42].

$$\rho_{nf} = (1 - \phi)\ \rho_{bf} + \phi\ \rho_{np} \quad (13)$$

$$C_{p,nf} = \frac{1}{\rho_{nf}}\left[(1 - \phi)\rho_{bf}C_{p,bf} + \phi\rho_{np}C_{p,np}\right] \quad (14)$$

$$\lambda_{nf} = \lambda_{bf}\left[\frac{\lambda_{np} + 2\lambda_{bf} + 2\phi\left(\lambda_{np} - \lambda_{bf}\right)}{\lambda_{np} + 2\lambda_{bf} - \phi\left(\lambda_{np} - \lambda_{bf}\right)}\right] \quad (15)$$

$$\mu_{nf} = A\mu_{bf}.e^{(B\phi)} \quad (16)$$

3. Optical simulations

Before developing a computational model of the fluid flow inside the HCE, a precise optical model is required for quantifying the heat flux of concentrated sun rays on the two concentric cylinders of the HCE. Accurate simulations are highly dependent on this heat flux as it is the main driving boundary condition. In this study, NREL's ray-tracing tool, SolTrace®, was used to generate the two circumferential profiles of heat flux while considering all sorts of optical errors based on Gaussian distributions. The mirrors were assumed to have slope and specularity errors of 3.5 and 0.5 mrad, respectively [6]. Fig. 1 depicts the intersection points of all 10^8 generated sunrays, as well as the paths of a few sample ones. The figure also shows the circumferential profiles of local concentration

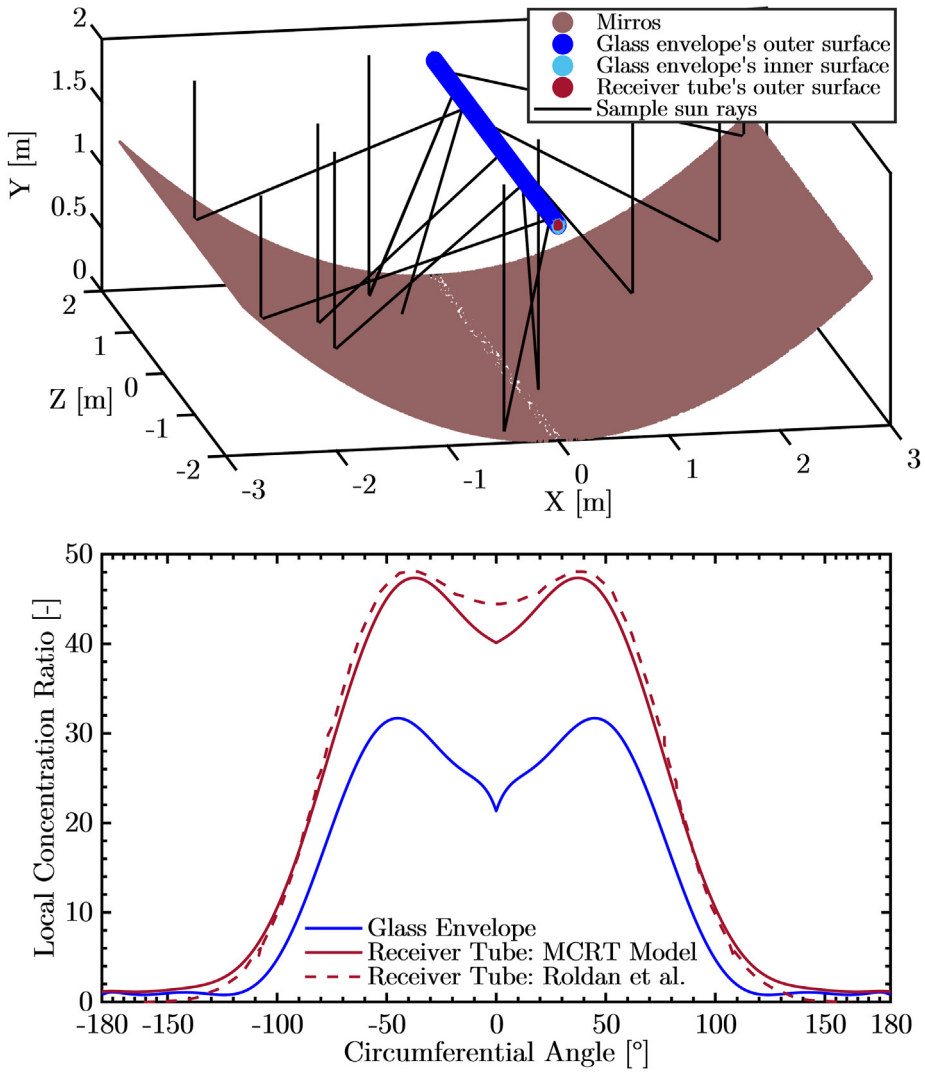


Fig. 1. Top: intersection points of all rays and paths of sample sun rays. Bottom: profiles of LCR on the glass envelope and receiver tube, compared to Roldan et al. [43].

ratio (LCR) on the two outer surfaces of the glass envelope and the receiver tube. By assuming the PTC to be facing the sun with a zero-incidence angle (dual-axis tracking), marginal variations of LCR along the HCE length were noticed. As shown in Fig. 1, the obtained profile of the receiver tube is in good agreement with the one described by Roldan et al. [43].

4. Computational flow simulations

4.1. Problem description

The studied problem comprises a nanofluid flow inside the absorber tube. The tube is subjected to external heat flux and loses heat to the surrounding air and sky enclosure through convection and radiation, respectively. The original problem comprises a 3D steady-state full-developed internal turbulent flow of incompressible fluid. The fluid flow rate is defined using the dimensionless Reynolds number (Re)

$$Re = \frac{4\,\dot{m}}{\mu_{nf}\,\pi\,d_t} \tag{17}$$

where $\dot{m}$ is the flow rate (kg/s) and d_t is the inner diameter of the absorber tube (m). Five values are considered for Re, namely: 4000, 10000, 20000, 30000, and 40000. These values align with the range tested by Dudley et al. [44] for a similar scale LS-2 PTC. Since the homogenous single-phase model is adopted, the flow is governed by the same transport equations of conventional fluids, except that the thermal properties of the fluid are adjusted using the aforementioned models in Section 6 to account for the presence of nanoparticles. The 3D conservation equations of mass, momentum, and energy are defined in the three cylindrical coordinates R, θ, and Z as follows [6].

Continuity

$$\nabla.\left(\rho_{nf} V\right) = 0 \tag{18}$$

Momentum

$$\nabla.\left(\rho_{nf} V V_i\right) = \frac{\partial p}{\partial X_i} + \nabla.\left(\mu_{nf} \nabla V_i\right) + S_i \tag{19}$$

Lastly, the energy equation

$$\nabla.\left(\rho_{nf} V C_{p,nf} T\right) = \nabla.\left(\lambda_{nf} \nabla T\right) \tag{20}$$

where the V is the velocity field $X = (R, \theta, Z)$, p is the local fluid pressure (Pa), T is the local fluid temperature (K), and S_i represents the remaining viscous terms in the momentum equation. For the three coordinates, this term is defined as

$$S_R = \rho_{nf} V_\theta \frac{V_\theta}{R} - \mu_{nf}\left[\frac{V_R}{R^2} + \frac{2}{R^2}\frac{V_\theta}{\partial\theta}\right] \tag{21}$$

$$S_\theta = -\rho_{nf} V_R \frac{V_\theta}{R} + \mu_{nf}\left[\frac{-V_\theta}{R^2} + \frac{2}{R^2}\frac{V_R}{\partial\theta}\right] \tag{22}$$

$$S_Z = 0 \tag{23}$$

For the closure of the aforementioned governing equations, the standard k-ε turbulent model was employed since it is widely used for handling boundary layer problems [6]. Another simplification for reducing the computational costs was made by reducing the 3D control volume to a 2D one using a 2D mesh. Hence, the aforementioned governing equations were solved by considering the variations of local fluid conditions in only longitudinal and radial directions. The following assumptions are made for defining the boundary conditions of the control volume [5,6,45]:

1. The external wall of the absorber tube is subjected to a constant heat flux boundary condition. This heat flux equals the mean of the original non-uniform heat flux of the 3D mesh.
2. There is no slip or penetration on all solid walls of the control volume.
3. A pressure outlet boundary condition was considered at the tube's outlet section.
4. A uniform velocity boundary condition was considered at the tube's inlet section.
5. A zero-gradient boundary condition was assumed at the two ends of the annular space since the low-pressure medium is stagnant and contained.
6. The convective heat transfer losses between the glass cover and the ambient air were determined using Newton's law of cooling. The corresponding convective heat transfer coefficient (h_o) was obtained based on the correlation of Mullick and Nanda: $h_o = w^{0.58} d_g^{-0.42}$ [46], where w is the mean normal wind velocity to the tube's length (taken as 3.0 m/s) and d_g is the outer diameter of the glass cover (m).
7. The radiative thermal losses were determined from the Steven-Boltzmann law by assuming a full view factor between the glass cover and the sky enclosure. The effective sky temperature (T_e) was considered as $T_e = 0.0552\, T_a^{1.5}$ [47], and the ambient air temperature was taken as 21.2 °C.
8. Inlet temperature was varied based on the following values 535.0, 625.0, 715.0, and 805.0 K. These values were considered as they all fall in the common operating range of the three molten salt types (see Section 2.2).

4.2. Implementation and validation

The CFD model was developed in ANSYS FLUENT® 19 R2. The following solution schemes and preferences have been adopted:

1 Special near-wall treatment.
2 SIMPLE algorithm for velocity-pressure coupling.
3 Second-order upwind finite difference approximation for the convective terms.
4 Central difference approximation for the diffusive terms
5 Convergence criteria of 10^{-8} (for energy equation) and 10^{-8} (for all other equations).

The aforementioned reduction of the control volume dimensionality is justified in Fig. 2(a). It is clear that the original 3D and the quasi 2D meshes are almost equivalent, with a maximum absolute deviation of less than 0.4%. The convergence and independence of the generated mesh were both checked through standard tests. Fig. 2(b) shows a comparison between three mesh sizes (coarse, medium, and fine) with 4673, 6016, and 11190 elements, respectively. As seen in this figure, the medium and fins meshes result in similar estimates of the system efficiency, with a maximum difference of less than 0.35%. Hence, the medium mesh was used in the rest of the study. Based on the standard grid convergence text, explained in Ref. [45], the maximum grid convergence index was lower than 0.5%, indicating the adequacy of the selected mesh size and configuration.

As shown in Fig. 2(c–d), the integrated MCRT-CFD model was validated against the well-known experiments of Dudley et al. [44] on LS-2 concentrator based on both the terminal temperature difference (the difference between outlet "T_o" and inlet "T_i" bulk fluid temperatures) and the thermal efficiency of the concentrator. It turned out that the maximum absolute errors are 9.7 and 8.3%, whereas the mean absolute errors are 8.3 and 5.5% for the two performance indices, respectively. Overall, a good agreement between the CFD model's estimations and the actual measurements that can be observed.

4.3. Performance indicators

In Sections 7.1 and 7.2, the thermo-hydraulic characteristics of the MSNFs will be discussed, whereas in Sections 7.3 and 7.4, the overall energetic and exergetic performances of the system will be analyzed. Accordingly, the results of the high-fidelity computational model are reduced to four key performance indices, namely: Nusselt number (Nu), friction factor (f), energy efficiency (η_{en}), and exergy efficiency (η_{ex}), all defined as follows

$$Nu = \frac{h_i\, d_t}{\lambda_{nf}} \tag{24}$$

$$f = \frac{2\, \Delta P\, d_t}{\rho_{nf}\, L\, v^2} \tag{25}$$

$$\eta_{en} = \frac{1}{I\, A_a}\left[\dot{m}\, C_{p,nf}\, (T_o - T_i) - \frac{\dot{m}\, \Delta P}{\eta_e\, \rho_{nf}}\right] \tag{26}$$

$$\eta_{ex} = \frac{\dot{m}\, C_{p,nf}\, (T_o - T_i) - \dot{m}\, C_{p,nf}\, T_a \ln\left(\frac{T_o}{T_i}\right) - \frac{\dot{m}\, T_a\, \Delta P}{T_m\, \rho_{nf}}}{I\, A_a\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right) + \frac{1}{3}\left(\frac{T_a}{T_s}\right)^4\right]} \tag{27}$$

where h_i is the convective heat transfer coefficient on the tube's inner surface (W/m^2.K), ΔP is the terminal pressure difference (Pa), L is the tube's length (m), v is the mean bulk fluid velocity (m/s), T_m is the mean bulk fluid temperature (K), I is the incident (non-concentrated) direct irradiance (fixed at 1000 W/m^2), A_a is the PTC's aperture area (m^2), η_e is a thermal-to-electric conversion efficiency (considered as 33.0% [5]) to estimate the thermal

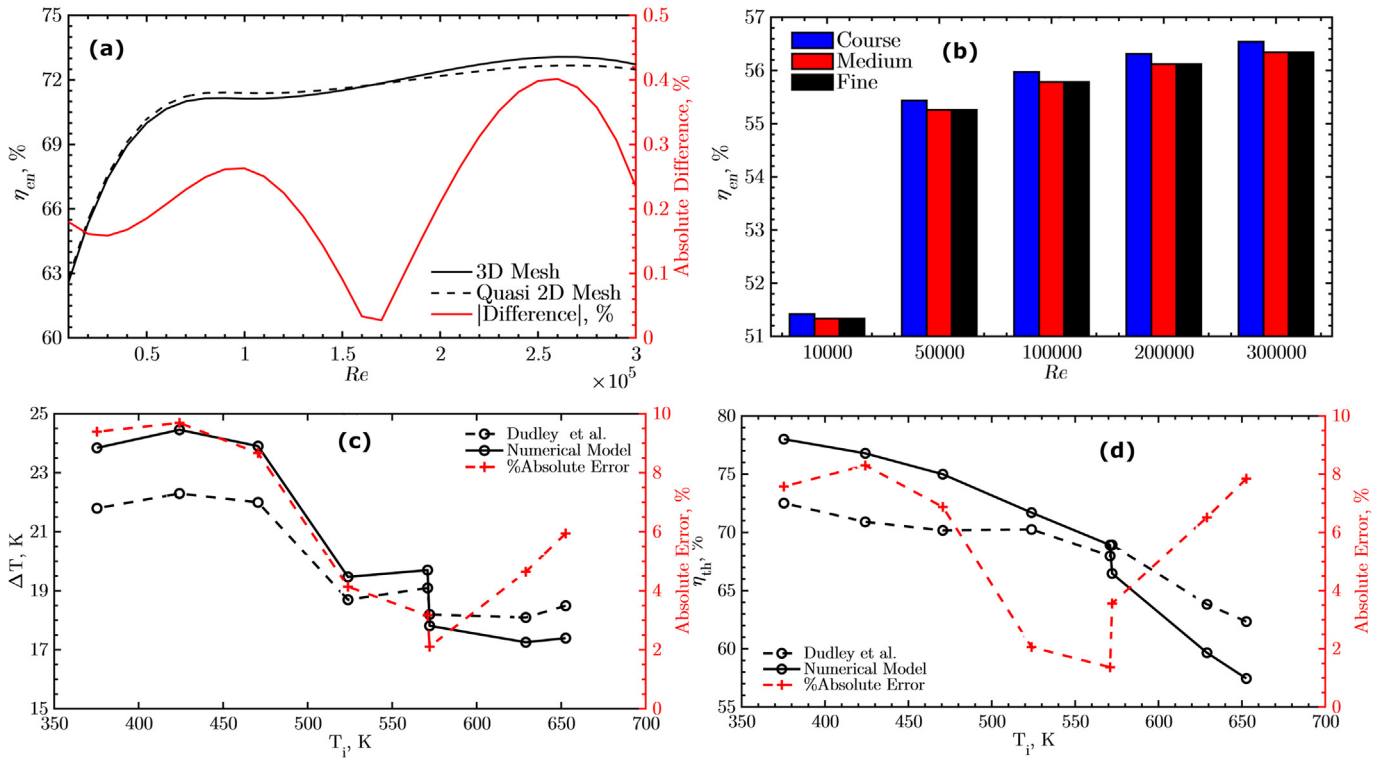


Fig. 2. (a) Comparison between 2D and 3D meshes in terms of overall energy efficiency, (b) grid independence tests, and (c–d) validation of the computational model against the experimental results reported by Dudley et al. [44].

equivalent of pumping power ($\dot{m}\,\Delta P/\rho_{nf}$), and T_s is the effective sun's surface temperature (considered as 5770 K) [5,6]. By excluding the equivalent pumping power in Equation (26), the thermal efficiency of the PTC, mentioned in Section 4.2, can be obtained.

5. Support Vector Regressors

As mentioned earlier, nine types of nanofluids have been studied. For each, three parameters have been altered, namely: ϕ (four values), T_i (four values), and Re (five values). The two later parameters have altered for the based fluids ($\phi = 0.0\,\%$) as well. This resulted in 780 different simulated cases. While this a sufficiently large number of cases to understand the enhancement mechanism of MSNF-based PTC, it is not large enough neither to fully map the system performance nor to train a precise model for optimization, especially when a heuristic optimization technique is utilized. This issue is addressed here by training a data-driven black-box model for replicating the results of the CFD simulations with drastically reduced computational expenses. In the present study, the support vector regressor (SVR) algorithm was utilized for this purpose. Two models were developed for the PTC's energy and exergy efficiencies.

Support Vector Machines (SVM) were initially developed by Vapnik for binary classification problems (Fig. 3(a)) [48]. Later, Support Vector Regressors (SVR) was developed for regression problems (Fig. 3(c)). In regression modeling, SVR constructs a tube with a diameter ε (i.e., the margin of tolerance) that best maps the input data to the output domain, as shown in Fig. 3(a). The goal is to find the flattest hyperplane ($w\,x + b = 0$) and the smallest tube diameter that contains most of the training data set, where w and b are the weights and biases of the model. This assures that the evaluated model compromises between accuracy (i.e., low bias) and generalization (i.e., low variance) [33,49,50], compared to the empirical optimization in more commonly used algorithms such as the multi-layered perceptron [6]. One of the unique features of SVM is its robustness against outliers. This is achieved by defining what is known as soft margin, which allows some points (i.e., noisy data or outliers) to lay outside the established tube. The impact of these outliers is defined using the loss function (ε-insensitive loss function in this study) and slack variables (ξ and ξ^*), shown in Fig. 3(d). The problem of finding the best tube that satisfies all of the aforementioned checks can be formulated based on the Karush-Kuhn-Tucker (KKT) optimization problem, which leads to a unique global optimum. For a linear problem, this optimization is formulated as

$$\min_{W,b,\xi}\left[0.5w^2 + C\sum_{n=1}^{N}\xi_n\right]$$

$$\textit{subject to}\ \begin{cases} Y_n(w\,x+b) \geq 1-\xi_n \\ \xi_n \geq 0 \end{cases} \tag{28}$$

where C stands for the box constraint, N is the size of the dataset, and Y is the target output of the model. Another feature of SVR is the mapping of the input data into a higher dimension domain before mapping them to the output variable (Fig. 3(d)). This is done by utilizing kernel functions to improve the prediction accuracy [51]. This can be done by a variety of mapping (kernel) functions such as linear, polynomial, and radial basis functions.

The search for the optimal kernel function and ε value for each model was automated using the Bayesian optimizer in a MATLAB® R2020 R1 environment while training the SVR models to achieve

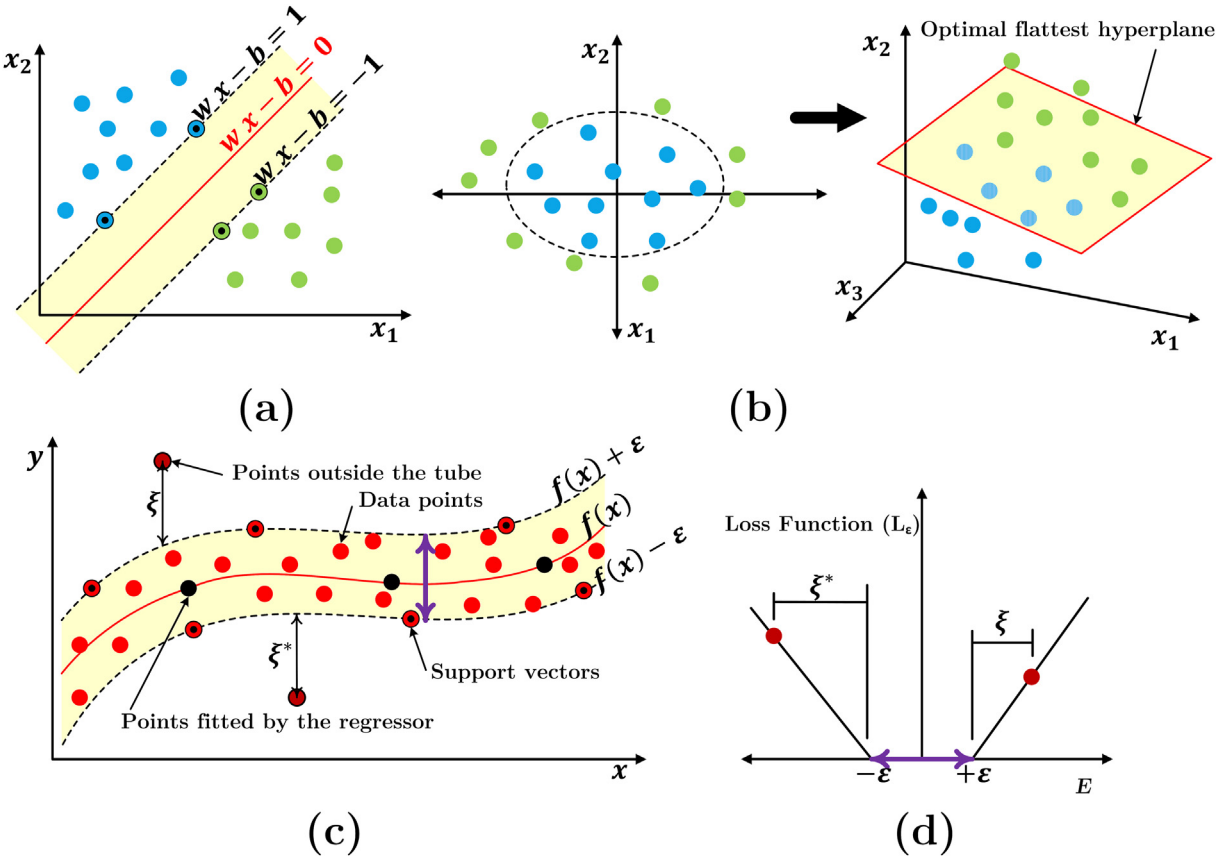


Fig. 3. (a) Binary support vector classifier, (b) higher-domain mapping, (c) support vector regressor, and (d) ε-insensitive loss function.

the highest possible accuracy. Accordingly, a fourth-order polynomial kernel with ε values of 0.141 and 0.2604 was selected for the models of energy and exergy efficiencies, respectively. Before the training process, data were randomized, and five-fold cross-validation was utilized. Fig. 4 shows that both models are showing an adequate representation of the data set generated from the MCRT-FVM model.

6. Optimization and decision-making techniques

Determining the optimal conditions is based on the decision maker's understanding of the problem and his/her intentions. The energy and exergy efficiencies were taken as responses to assess the performances of the different nanofluids. The higher both efficiencies are, the better the performance is. In the present study, a total of 12 combinations of molten salts and nanoparticles (nine nanofluids and three base fluids) were optimized individually to select the optimum operation conditions, namely, nanoparticle loading, temperature, and Reynolds number) for each of them. Therefore, the optimization problems of the base fluids and nanofluids in the current study can be expressed as in Equations (29) and (30), respectively.

$$\begin{cases} Maximize\ \eta_{en} = f_1(T_i, Re) \\ Maximize\ \eta_{ex} = f_2(T_i, Re) \\ Subject\ to: \\ 535 \le T_i \le 805\ K \\ 4\times10^3 \le R_e \le 40\times10^3 \end{cases} \quad (29)$$

$$\begin{cases} Maximize\ \eta_{en} = f_1(\phi, T_i, Re) \\ Maximize\ \eta_{ex} = f_2(\phi, T_i, Re) \\ Subject\ to: \\ 0.0 \le \phi \le 4.0\% \\ 535 \le T_i \le 805\ K \\ 4\times10^3 \le R_e \le 40\times10^3 \end{cases} \quad (30)$$

Besides, two other optimization problems were also considered, as represented in Equations (31) and (32)

$$\begin{cases} Maximize\ \eta_{en} = f_1(\phi, T_i, Re) \\ Maximize\ \eta_{ex} = f_2(\phi, T_i, Re) \\ Subject\ to: \\ 0.0 \le \phi \le 4.0\% \\ T_i = 805\ K \\ R_e = 4\times10^3 \end{cases} \quad (31)$$

$$\begin{cases} Maximize\ \nabla\eta_{en}\% = f_1(\phi, T_i, Re) \\ Maximize\ \nabla\eta_{ex}\% = f_2(\phi, T_i, Re) \\ Subject\ to: \\ 0.0 \le \phi \le 4.0\% \\ T_i = 805\ K \\ R_e = 4\times10^3 \end{cases} \quad (32)$$

Where $\nabla\eta\%$ is the percentage enhancement in the system efficiency (the subscript indicates the efficiency type). This percentage is calculated as shown in Equation (33), where η_{NF} and η_{BF} are system efficiencies corresponding to the nanofluid and its base fluid, respectively, both evaluated at the same operating conditions.

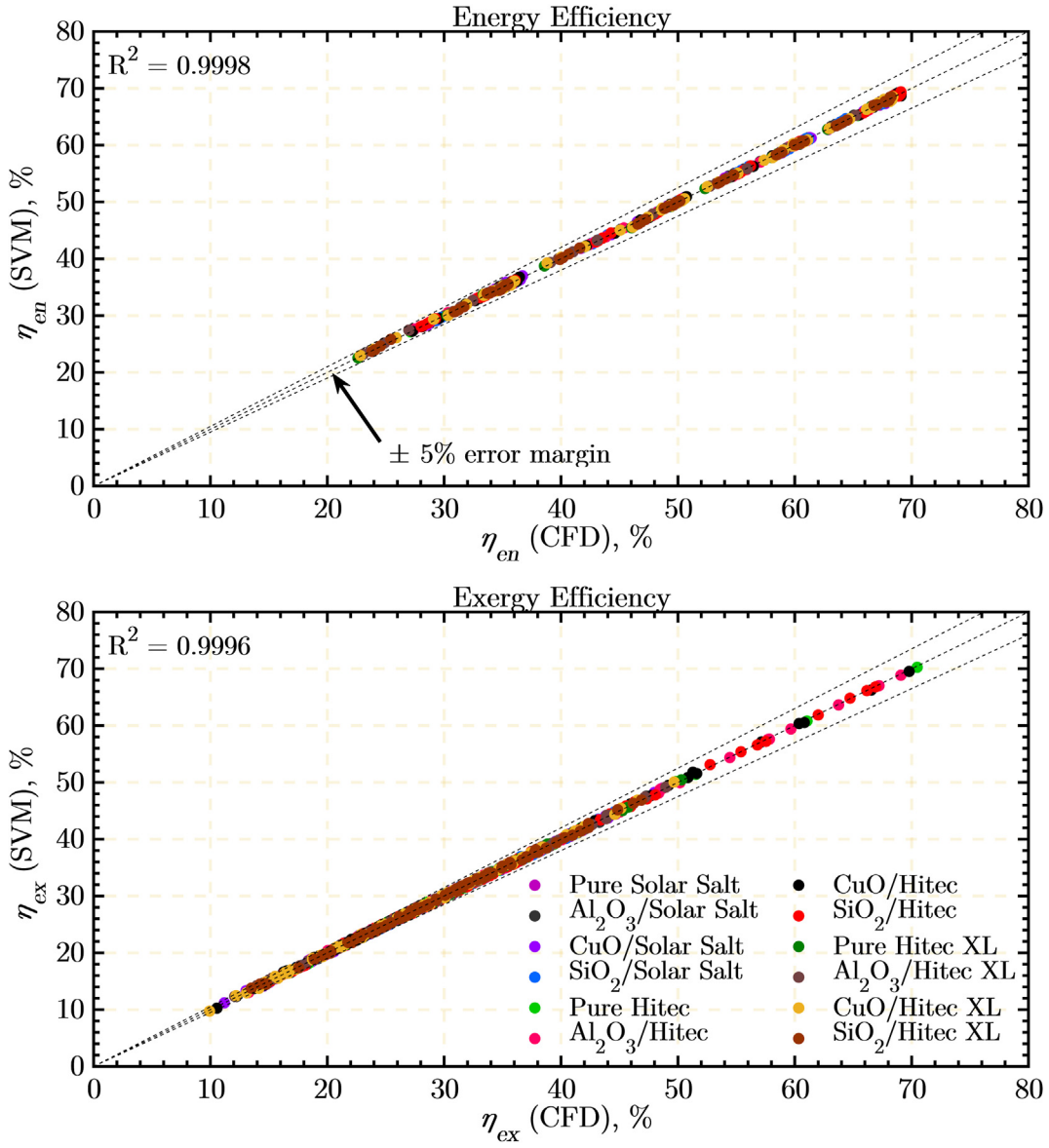


Fig. 4. Agreement between CFD simulation results and corresponding values obtained using the reduced-order SVM model.

$$\nabla\eta\% = \frac{\eta_{NF-\eta_{BF}}}{\eta_{BF}} * 100 \tag{33}$$

The SVM models described in Section 5 were used as the objective functions for both exergy and energy efficiencies. Afterward, a heuristic particle swarm optimizer (PSO) integrated with a non-dominate-sorting genetic algorithm (NSGA) was applied. The reason for coupling PSO and NSGA is that while PSO is a fast and efficient optimization technique, it is still a single-objective optimizer. The dual objective nature of the problem of interest necessitates the utilization of NSGA [52,53]. The basic concept of the hybrid PSO-NSGA optimizer is illustrated in Fig. 5. The PSO algorithm mimics the strategy followed by a swarm of birds while searching for food. Starting with an initial population of particles (i.e., food seeker), each particle in the swarm is locally searching in its proximity for food (i.e., positions in the search domain that best satisfies the optimization criteria). Every while (i.e., iteration/generation), these particles communicate with each other, sharing real-time updates on hot spots (locally optimum points). Each particle in the swarm builds its own search strategy based on both its personal experience (i.e., self-confidence) and the information shared among the swarm individuals (swarm-confidence factor). Two factors are used to determine the dependency on these two sources of information when planning a future research strategy, mainly C1 (self-confidence factor) and C2 (swarm confidence factor). In this study, these factors were assigned equal weights. The problem with using PSO solely to solve the optimization problem lies in the difficulty of defining the hot spots mentioned above. This is due to the intrinsic feature of multi-objective optimization (MOO), namely the conflict between the objectives. As shown in Fig. 5, NSGA is utilized to define the global optimum so that the individual particles can follow a direction that considers it. These concepts always depend on comparing the particle with the current population to find out the non-dominated solutions (particles), also known as the Pareto front solutions. These solutions are then saved in a repository. Afterward, predetermined criteria are utilized for the selection of global optimum from the repository [54,55]. This cycle repeats itself until no significant change is noticed in the

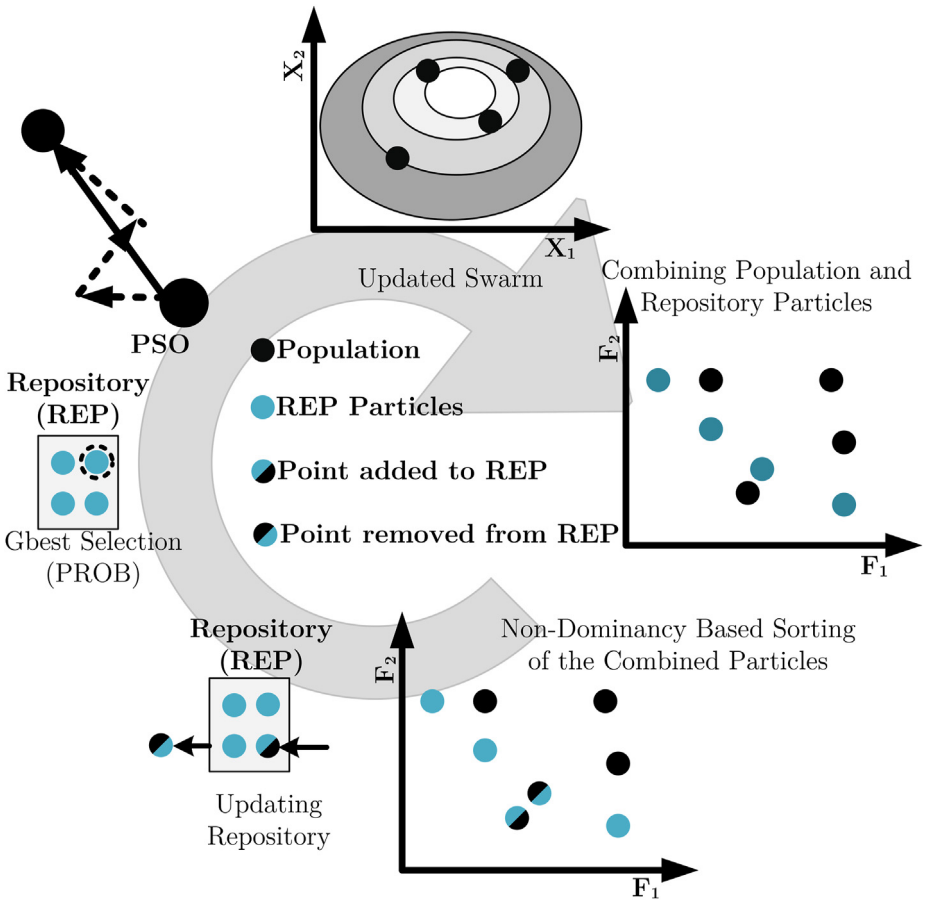


Fig. 5. PSO-NSGA multi-objective optimization algorithm.

Pareto front or for a predetermined maximum number of generations.

The above-mentioned procedures are repeated for each of the working fluids under investigation, and a separate Pareto front is evaluated for each fluid. To compare the different fluids, these Pareto fronts are combined, and -again- the NSGA is applied to find the global Pareto front of all fluids. At this point, the Pareto front offers the designer a unique set of solutions that outperform all the other tested solutions. However, in this unique set, no solution outperforms the others, i.e., all solutions are non-dominated. For this reason, the Technique of Order Preference Similarity to the Ideal Solution (TOPSIS) represents a systematic way to apply the designer preference for the final selection. TOPSIS is a distance-based method that ranks the solutions based on their closeness to a utopian imaginary solution point (i.e., optimizing both objectives) and their remoteness from an imaginary dystopian point [56]. Based on these two distances, the points are ranked. Points that are respectively the closest to and furthest from the utopian and dystopian points are highly ranked. It is worth mentioning that TOPSIS (and its alternatives) by no means is a substitute for the domain of expertise. In fact, TOPSIS is built on the idea of transforming the descriptive importance to a quantitative weight that can be applied to find the most suitable solution from the Pareto front. Here, equal weights were given for both objectives (energy and exergy efficiencies).

7. Results and discussion

In the following sections, detailed discussions of the performance of a PTC operating with MSNFs are provided. These discussions cover the performance from thermal (Section 7.1), hydraulic (Section 7.2), energy (Section 7.3), and exergy (Section 7.4) perspectives based on the results of the MCRT-FVM model. The full set of results are provided in Appendix A. Afterward, Section 7.5 provides the results of the system optimizations.

7.1. Thermal performance analysis

Adding nanoparticles to the different considered molten salts has a positive impact on the thermal properties of the resultant nanofluids, which consequently improves the internal heat removal rate, presented in terms of *Nu*. Fig. 6 depicts the variations of *Nu* as *Re* increases for all MSNF types and volume concentrations at $T_i = 805$ K. The results of all modified fluids show an increasing *Nu* in a semi-linear trend as *Re* increase. Such a behavior aligns with reported studies for other types of nanofluids, e.g., Refs. [6,8]. The maximum *Nu* depicted in Fig. 6 (535.5) is obtained using CuO-Hitec XL nanofluid at a volume concentration of 4.0% and *Re* of 4×10^4. However, the maximum *Nu* obtained among all 780 simulated cases is 892.3, which is obtained using CuO-Hitec nanofluid at $\phi = 4.0\%$, *Re* of 40×10^4, and T_i of 535 K. On the other hand, the minimum *Nu* obtained among all simulated cases is 20.46, which is obtained using pure Hitec at *Re* of 4×10^4 and T_i of 805 K. Higher *Nu* can be achieved by operating the PTC at lower temperatures and higher flow rates since the heat removal rate is directly proportional to both the flow rate and the tube-to-fluid temperature difference. The same trend is noticed when increasing the nanoparticle loading at fixed *Re* values, except that considerable

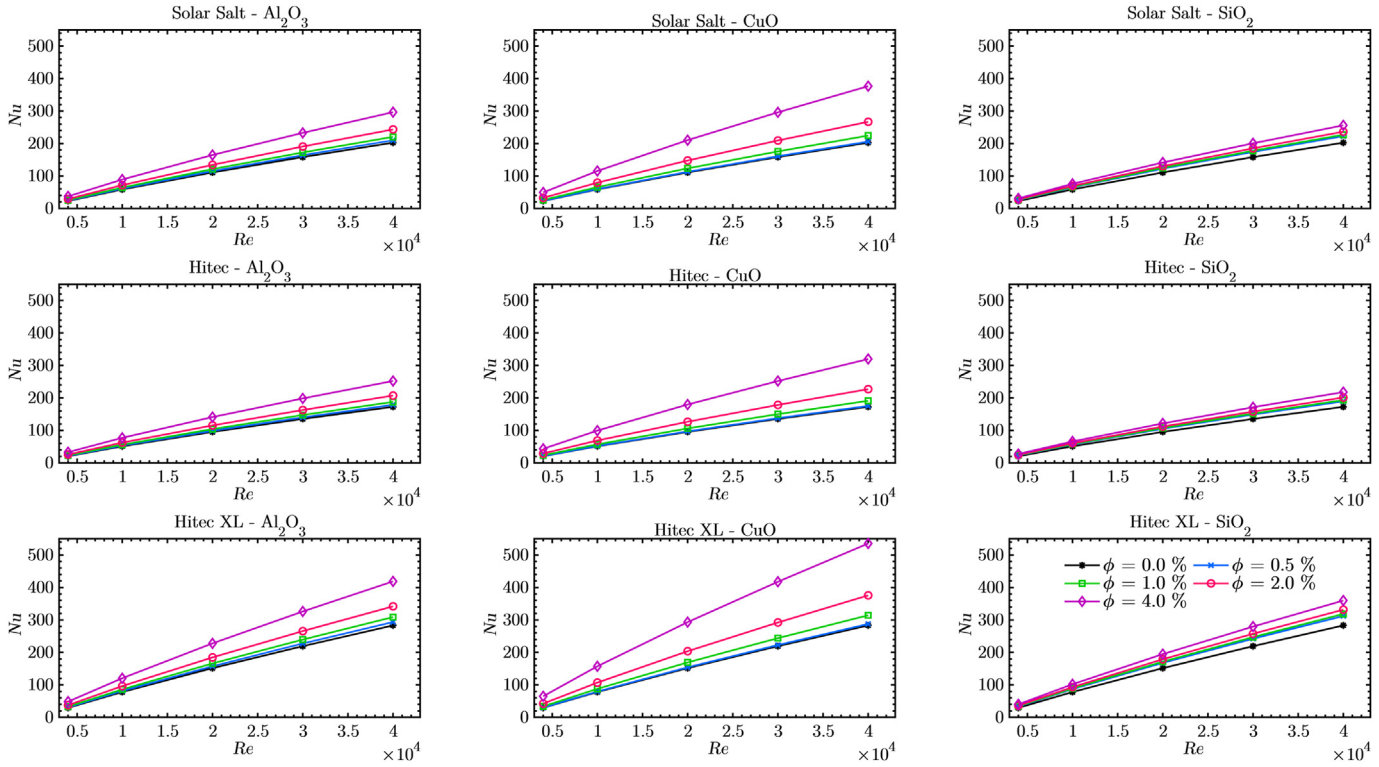


Fig. 6. Variations in internal Nusselt number with the increase in Reynolds number.

non-linearities are noticed for CuO nanofluids, which can be attributed to the higher corresponding B coefficient in the viscosity equation (Section 2.2). The relative performance can be assessed by examining the variations in the Nusselt number ratio for all MSNFs at different concentrations and Reynolds numbers. The Nusselt number ratio is simply the ratio between *Nu* of the MSNF and that of the corresponding pure salt (Nu_0). At the same temperature, there is a sharp decrease in *Nu* ratio at the beginning of the studied *Re* range, especially at high volume concentration. Then, the decrease becomes slight and *Nu* almost stabilizes as *Re* increase further. The reason is that at higher *Re* values, the heat transfer rate is mainly driven by the linear and angular momentums of the fluid, resulting in higher rates of fluid mixing rather than the improved thermal properties or the Brownian motion of the nanoparticles. Besides, the operating temperature has a lesser impact on the *Nu* ratio, compared to *Re*. Among the different simulated cases, the maximum *Nu* ratio is 2.218, which is obtained using CuO-Solar Salt nanofluid at $\phi = 2.0\%$, *Re* of 4×10^4, and T_i of 805 K. Meanwhile, the minimum *Nu* ratio is 1.016, which is obtained using CuO-Solar Salt nanofluid at $\phi = 2.0\%$, *Re* of 40×10^4, and T_i of 535 K.

7.2. Hydraulic performance analysis

The absolute hydraulic performance of the PTC in terms of Darcy's friction factor is depicted in Fig. 7 for T_i of 805 K. The results depict a decreasing friction factor as the Reynolds number increases. Moreover, the friction factor increases with increasing the volumetric concentrations of nanoparticles for all considered MSNFs. This is driven by the amplified viscosity of the nanofluid compared to its base salt (equation (16)). A more notable increase in the friction factor is associated with CuO nanoparticles, regardless of the molten salt type, whereas minimal increases in the friction factor are associated with SiO_2-based MSNFs. This is directly correlated to the corresponding coefficients of the nanoparticles in equation (16). The variations in the friction factor ratio with *Re* number can be examined using the results provided in Appendix A. Again, this is the ratio between the friction factor of the MSNF and the friction factor of the corresponding pure salt (f_0). At higher *Re* values, the friction factor ratio stabilizes since the pressure drop is mainly driven by the high flow rate, rather than the increasing fluid viscosity. Both absolute and relative friction factors increase as the temperature decreases to 535 K since the fluid's viscosity increases. However, this increase is marginal, and -again- the operating flow rate is far more influential. Among the different simulated cases, the maximum f is 0.139, which occurs when operating using CuO-Hitec nanofluid at $\phi = 2.0\%$, *Re* of 4×10^4, and T_i of 535 K. Meanwhile, the minimum f is 0.024, achieved using pure Hitec at *Re* of 40×10^4 and T_i of 805 K. On the other hand, the maximum f ratio is 4.434, which occurs when operating using Al_2O_3-Solar Salt nanofluid at $\phi = 4.0\%$, *Re* of 4×10^4, and T_i of 805 K. Meanwhile, the minimum f ratio is 1.036, achieved using pure CuO-Solar Salt at $\phi = 2.0\%$, *Re* of 40×10^4, and T_i of 535 K.

7.3. Energetic performance analysis

Fig. 8 shows the variations in the energy efficiency of PTCs, operating with various molten salts and MSNFs, with *Re* at an inlet temperature of 805 Ks. It can be deduced from the figure how the energy efficiency trends to increase with increasing *Re* and volumetric concertation. Those results are attributed to the aforementioned enhanced thermal energy removal and reduced thermal losses from the receiver tube's outer surface. At lower operating temperatures, higher efficiencies than those in Fig. 8 can be obtained. Specifically, at the highest considered *Re* (40000) and lowest temperature (535 K), the energy efficiency approaches a limit of ~70.0% for all types of fluids. At these conditions, the impact of nanoparticle concentration is minimal since the performance is mainly improved by the lower system temperatures and higher

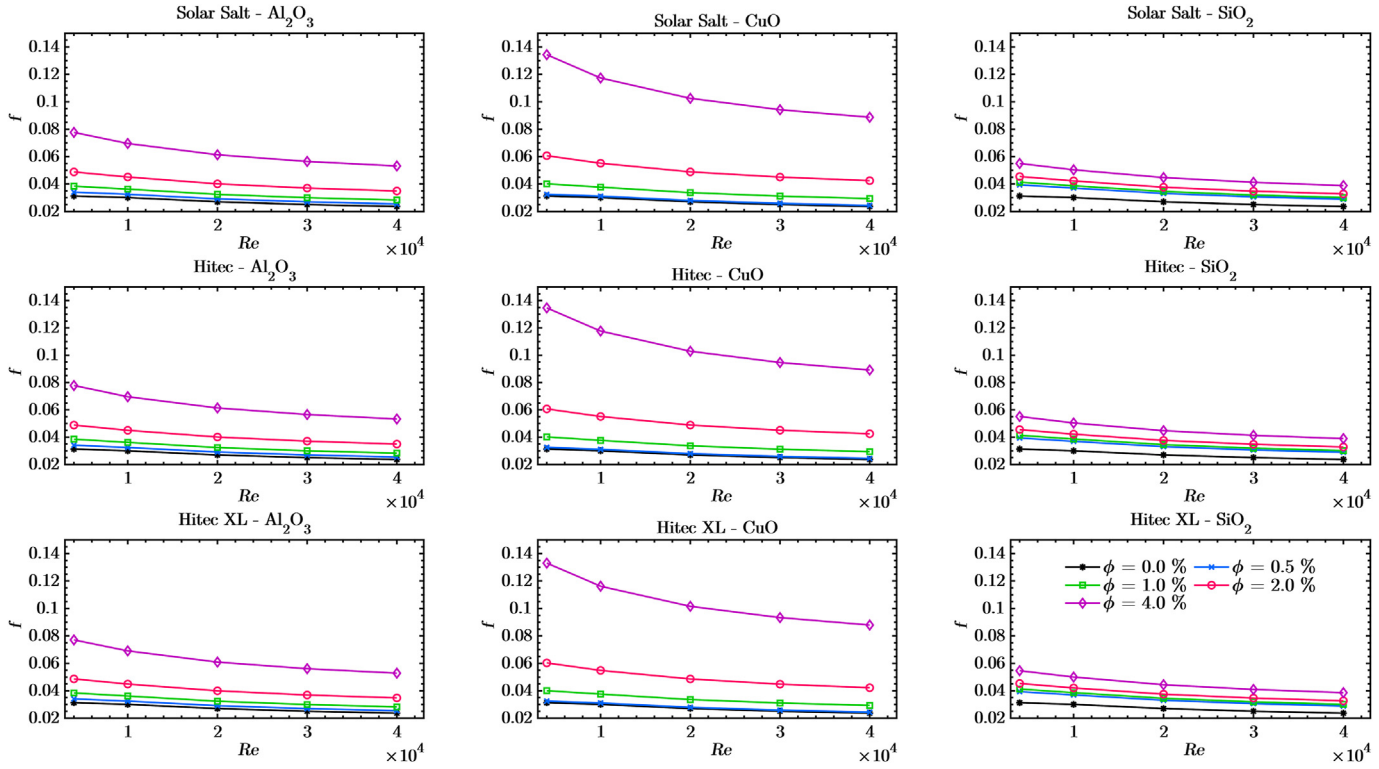


Fig. 7. Variations in friction factor with the increase in Reynolds number.

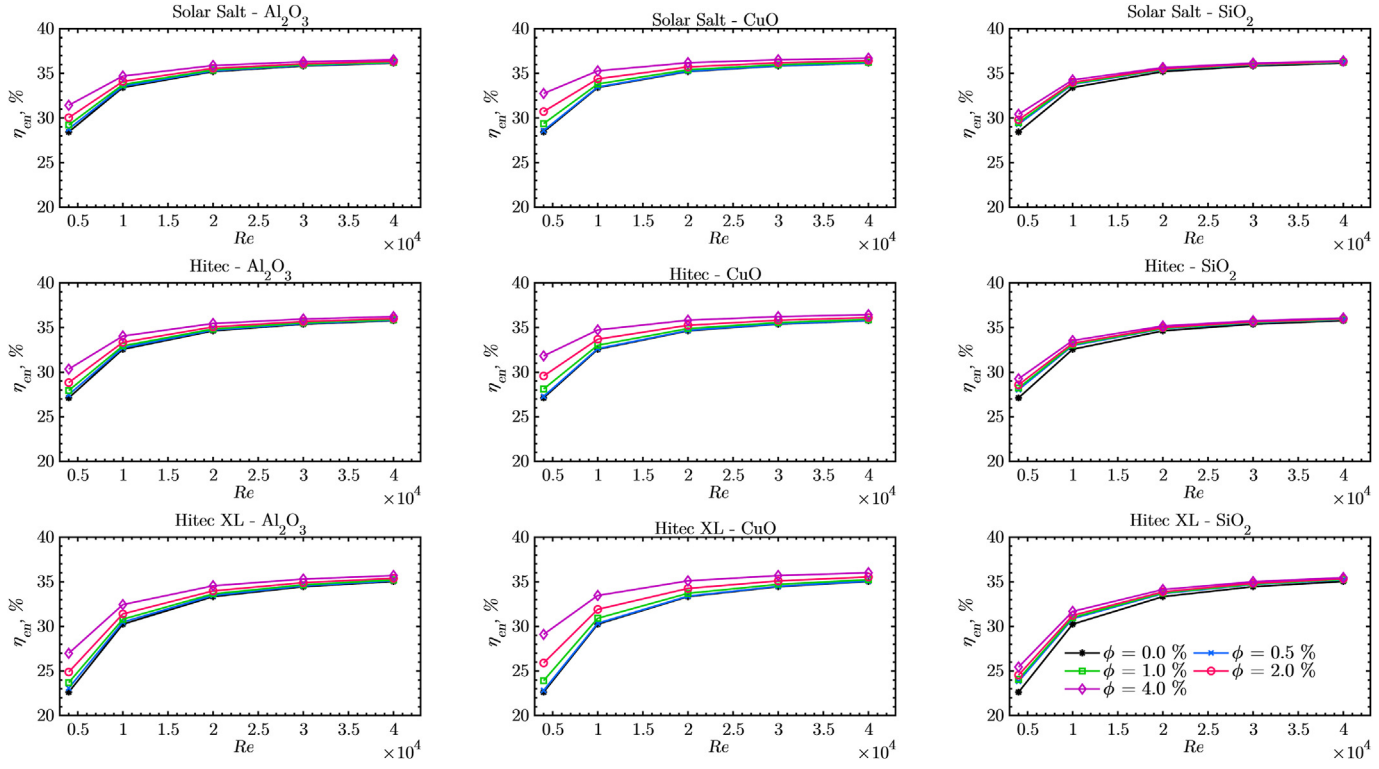


Fig. 8. Variations in energy efficiency with the increase in Reynolds number.

flow rates. In other words, the advantage of adding nanoparticles is more pronounced at low flow rates. In such a case, increasing concentration consistently improves overall energy efficiency. The efficiency also decreases as the operating temperature increases due to the intense thermal losses at higher temperatures (as typically presented in the efficiency curves of solar collectors). At this

point, it is worth mentioning that the efficiency of the PTC operating with CuO-Hitec of high concentrations starts decreasing again, rather than stabilizing, as *Re* increase beyond ~30000 for low operating temperatures. This is because at that point, the pumping power consumption increases drastically due to the increased MSNF's viscosity. Such a point was reported by the authors [6] and others, and it can be more significant if a wider *Re* range was considered. Among the different simulated cases, the maximum η_{en} is 69.089%, achieved when operating using SiO_2-Hitec nanofluid at $\phi = 1.0\%$, *Re* of 40×10^4, and T_i of 535 K. Meanwhile, the minimum η_{en} is 22.628%, which occurs when using pure Hitec XL at *Re* of 4×10^4 and T_i of 805 K.

To examine the potential enhancements in PTC's energetic performance when replacing conventional salts with MSNFs, a new parameter (η_{en}) is used. This parameter presents the percentage difference between the energy efficiency when using a nanofluid and that when using the corresponding pure molten salt. To fully map these potential enhancements, the precise SVR model was used to simulate 150000 different combinations of salt type, nanoparticle type, concentration, temperature, and Reynolds number, all within the considered range of the CFD simulations (to avoid overfitting). The results are shown in the contour plots in Fig. 9. In this figure, only regions with achieved enhancements are shown. The blank areas represent combinations of fluid type and operating conditions where the addition of nanoparticles would have an adverse impact on the PTC's energy efficiency. Solar salt offers the largest range of operating conditions where adding nanoparticles is effective. Yet, the expected enhancement is mostly marginal, except for very high concentrations (around 4.0%) and especially at high temperatures and low flow rates, where percentage enhancements up to 17.0% can be achieved. This holds for Hitec-based fluids, except that narrower ranges of operating conditions can be considered for enhancing the performance. At medium temperatures and high flow rates, no enhancement can be achieved. Besides, the levels of enhancements are generally higher than those of Solar Salt. Finally, Hitec XL corresponds to a very limited operation range where performance enhancements can be achieved. Even in such ranges, the expected enhancements are less considerable, compared to Solar Salt and Hitec. When utilizing this figure, it is recommended to avoid dark blue regions that are very close to the white areas for two reasons: 1) the enhancement is marginal (slightly higher than 0.0%), and 2) to avoid propagated numerical errors in both CFD model and SVR.

7.4. Exergetic performance analysis

Evaluations of the exergetic efficiency of PTCs operating with various salts and MSNFs have been performed, as demonstrated in Fig. 10. The figure presents the exergetic performance response of the PTC as *Re* increases for all volumetric concentrations, but at a constant temperature of 805 K. The figure demonstrates relatively more diverse responses as the design and/or operating conditions change compared to those of the energetic performance. The exergy efficiency consistently increases by increasing *Re* since this decreases heat loss to the environment, thus the associated heat transfer-related irreversibilities. By operating at lower temperatures (e.g., 535 K), higher exergy efficiencies, up to ~70.0%, can be achieved due to the reduced thermal losses to the surrounding air and sky enclosure. Higher heat loss rates represent an additional source of entropy generation (i.e., exergy destruction). For all salt types and operating conditions, increasing the nanoparticles' concentrations reduces the exergy efficiency. This is attributed to the corresponding increases in both friction-related irreversibilities and dynamic viscosity of the nanofluid. Among the different simulated cases, the maximum η_{ex} is 70.477%, achieved when operating using pure Hitec at *Re* of 40×10^4 and T_i of 535 K. Meanwhile, the minimum η_{ex} is 9.923%, achieved using CuO-Hitec XL at ϕ of 4.0%, *Re* of 4×10^4, and T_i of 805 K.

Fig. 11 shows similar heat maps for the percentage enhancement in exergy efficiency. It can be seen that almost only Hitec-based MSNFs have a positive impact on the exergetic performance. This -again- takes place in a very limited operating range of high flow rates and low temperatures and for low nanoparticle concentrations, where percentage enhancements up to 42.0% can be

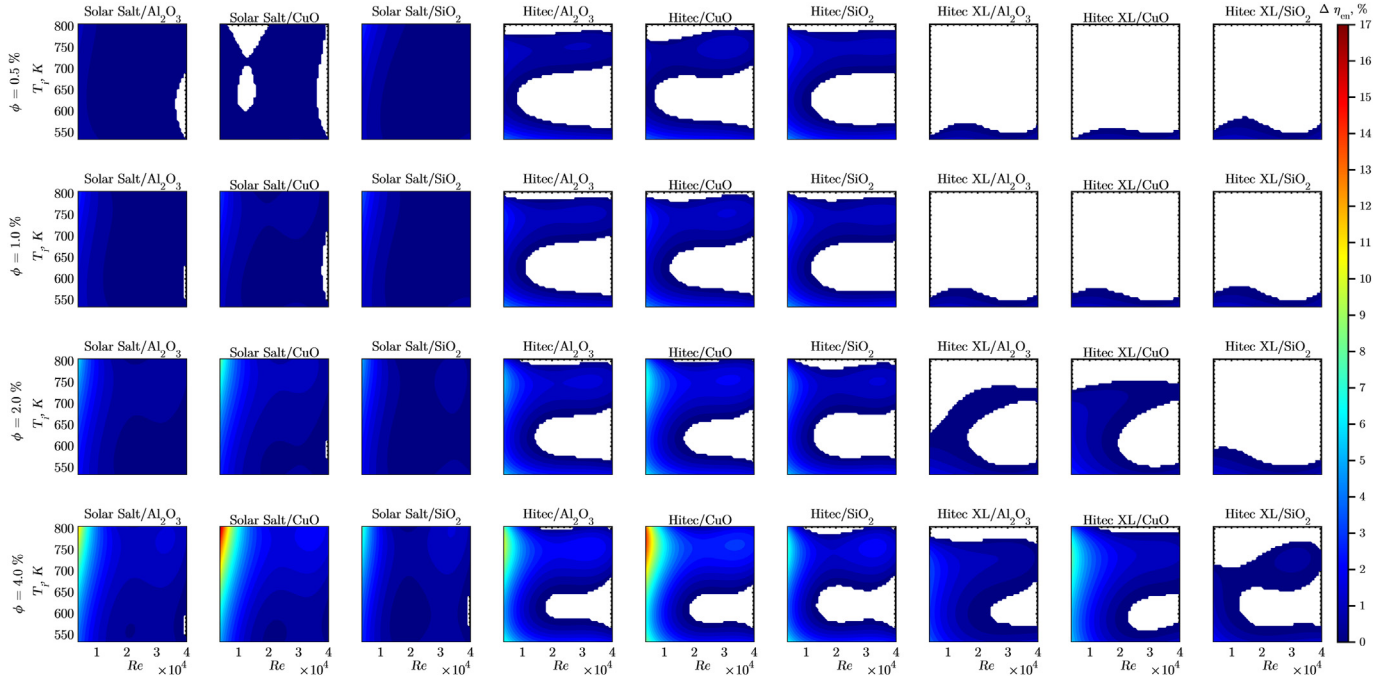


Fig. 9. Performance enhancement map in terms of energy efficiency. Blank white areas correspond to an adverse impact of the nanoparticles.

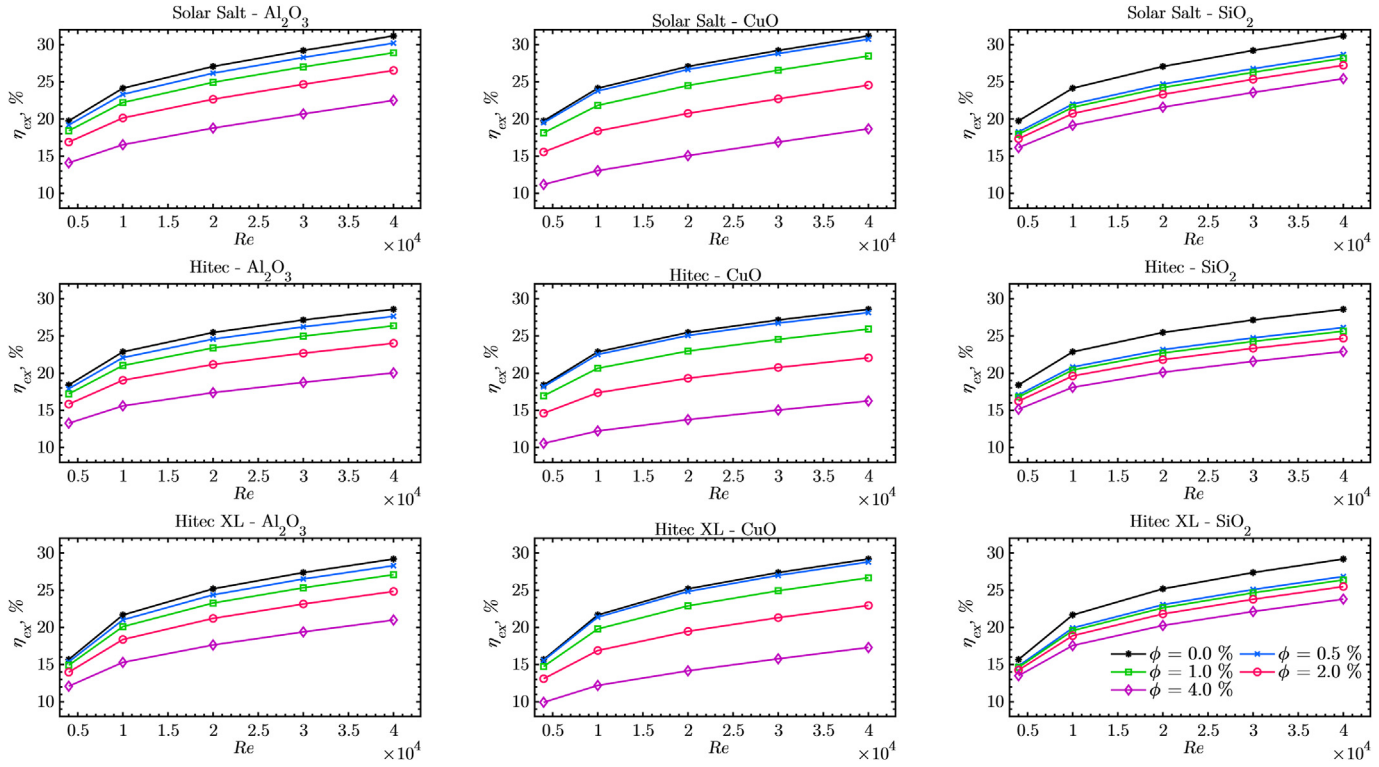


Fig. 10. Variations in exergy efficiency with increase in Reynolds number.

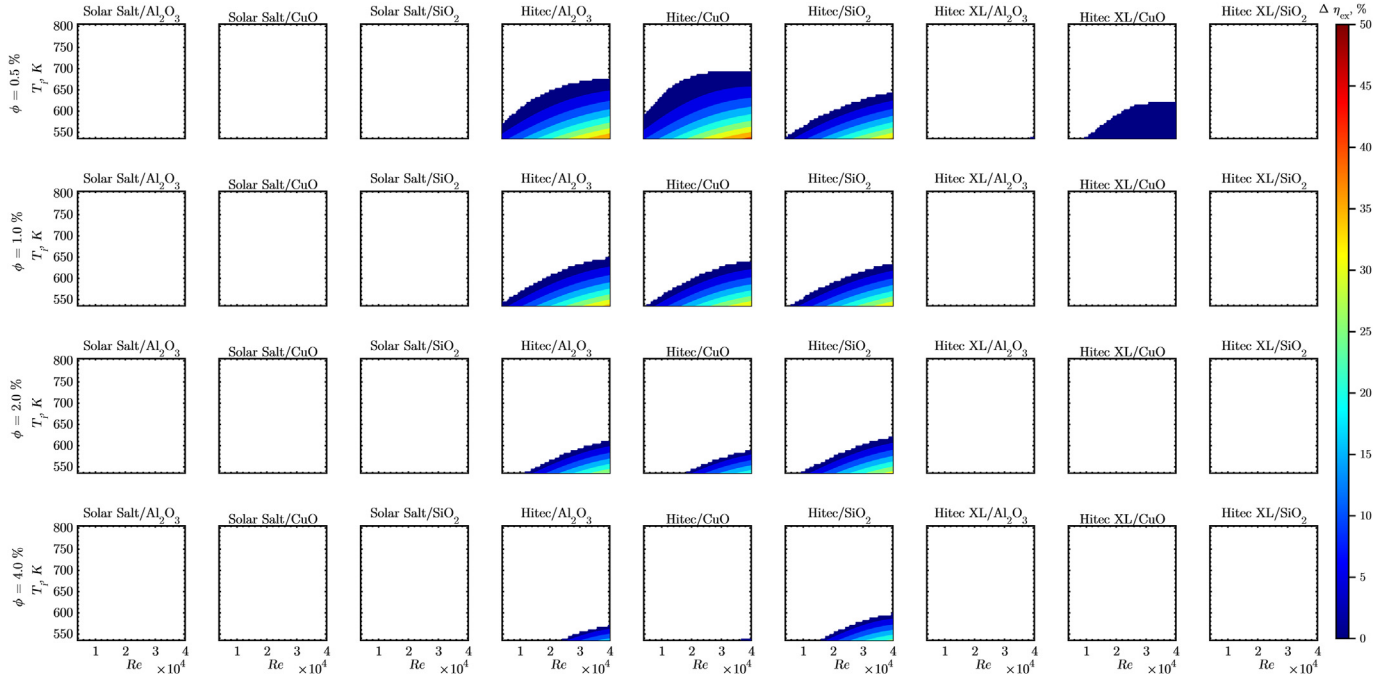


Fig. 11. Performance enhancement map in terms of exergy efficiency. Blank white areas correspond to an adverse impact of the nanoparticles.

achieved.

7.5. Optimal design and operation

In this section, optimization results based on the methodology outlined in Section 6 are discussed. The obtained results for each working fluid are shown in Fig. 12(a). It is evident that the Pareto fronts of the different working fluids differ considerably. In general, CuO/Hitec outperforms all other nanofluids based on both energy and exergy efficiencies (see the individual Pareto fronts in Fig. 12(a)). Even more, it is noticeable that Hitech-based MSNFs generally show better performances compared to both Solar Salt-

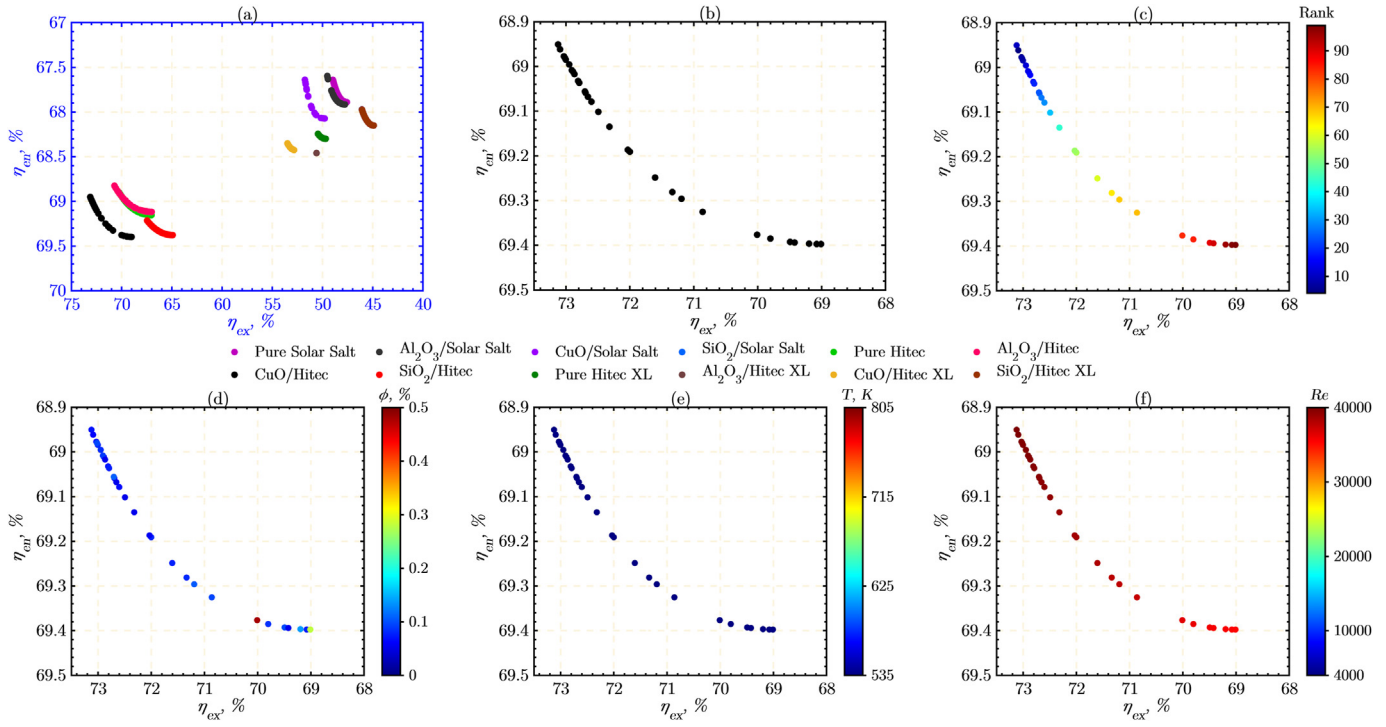


Fig. 12. Results of the multi-criteria optimization of absolute performance. (a) Individual Pareto fronts, (b) global Pareto front, (c) TOPSIS-based ranks of the global Pareto solutions, (d)–(f) global Pareto solutions mapped based on the volumetric concentration, fluid inlet temperature, and Reynolds number, respectively.

and Hitech XL-based ones. Fig. 12(c) shows the TOPSIS-based ranks of the global Pareto solutions (CuO/Hitec). The lower the rank number, the better the solution in terms of both efficiencies combined. It is important to note that these solutions depend on the weights assigned to each objective by the decision-maker, i.e., different weights will indeed lead to different ranks. Based on equal weights, the optimum final solution achieved energetic and exergetic efficiencies of 73.071 and 68.96%, respectively. This corresponds to T_i, *Re*, and ϕ of 535 K, 39912.98, and 0.0619%, respectively. Note that these are not the maximum energy and exergy efficiencies, but the energy and exergy efficiencies corresponding to the best solution of the multi-criteria optimization. The highest values of the two efficiencies are respectively 73.14 and 69.39%. These two extreme values represent the optimum solutions of single-objective optimizations based on energy and exergy efficiency, respectively. For energy efficiency, this was achieved at T_i, *Re*, and ϕ of 535 K, 35520, and 0.139% respectively, while for the exergy one, this was achieved at T_i, *Re*, and ϕ 535 K, 39943, 0.0619% respectively.

Fig. 13(a) shows a comparison between three key solutions, mainly the solution with the highest energy efficiency, highest exergy efficiency, and best TOPSIS rank. Notably, all of these solutions occur at low T_i, high Re, and CuO/Hitec with low nanoparticle concentration with a maximum-recorded concentration of 0.139%, which is considerably low relative to the high-end of the studied domain (4.0%). The reason why nanofluids are not appearing with any of these solutions is as follows. At low T_i and high *Re*, PTC's thermal efficiency is already at its peak. In other words, thermal losses are already limited due to the high rate of heat removal at the inner side of the tube (due to the high flow rate) and the high-temperature difference between the tube's surface and the fluid (low T_i). Increasing the loading of nanoparticles cannot further enhance the performance (e.g., reduce the optical loss) and may even lead to adverse effects, i.e., increasing the pressure drop. Fig. 12(d–f) shows the same results in Fig. 13(a) but for all Pareto front points, and similar conclusions can be drawn. The key takeaway from this discussion is that the use of nanofluids at certain conditions may not be justified from an energy perspective, let alone the economic aspect.

However, it is critical to notice that nanofluids' utilization under specific conditions may be of great advantage. To illustrate this, Figs. 13(b) and Figure 14 consider the special case of PTC operation under high temperatures and low flow rates. This is demonstrated in the two figures for $T_i = 805$ K and low $Re = 4000$. Compared to Fig. 12, the presence of nanofluids is more significant in the final Pareto front. The nanoparticles' concentration of the top-ranked solution is 2.23%, compared to 0.0619% if the whole domain was considered. To show more insights in this regard, the two radar plots in (Fig. 13 (a) and (b)) are compared. For the solution of maximum energy efficiency, the optimum nanoparticles' concentration is 0.335%, instead of the aforementioned 0.1386%. Interestingly, for Figs. 13(b) and Figure 14, the working fluid with maximum exergy is pure solar salt, achieving the highest value of 28.048%. The fact that pure solar salt with nanofluid achieved the highest exergy indicates that if the main concern of the decision-maker is to maximize the exergy efficiency at high temperature and low Re, the idea of adding nanoparticles to the molten salt should be ditched. This also has been pointed out in the discussion of Fig. 10.

Another point that is worth considering in the optimization problem is the relative enhancement in both efficiencies when the molten salt is replaced with an MSNF. This is useful as it masks the effect of the base fluid. It is evident in Fig. 12 that the Hitech molten salt outperforms the other fluids. It is even noticeable that pure Hitech salt also outperforms other MSNFs based on Solar Salt and Hitec. While the performance of the PTC can be easily boosted by replacing Solar Salt and Hitec XL with Hitec, other considerations must be involved, such as the environmental impact, availability of the salt type in the local market, and -most importantly-the costs of

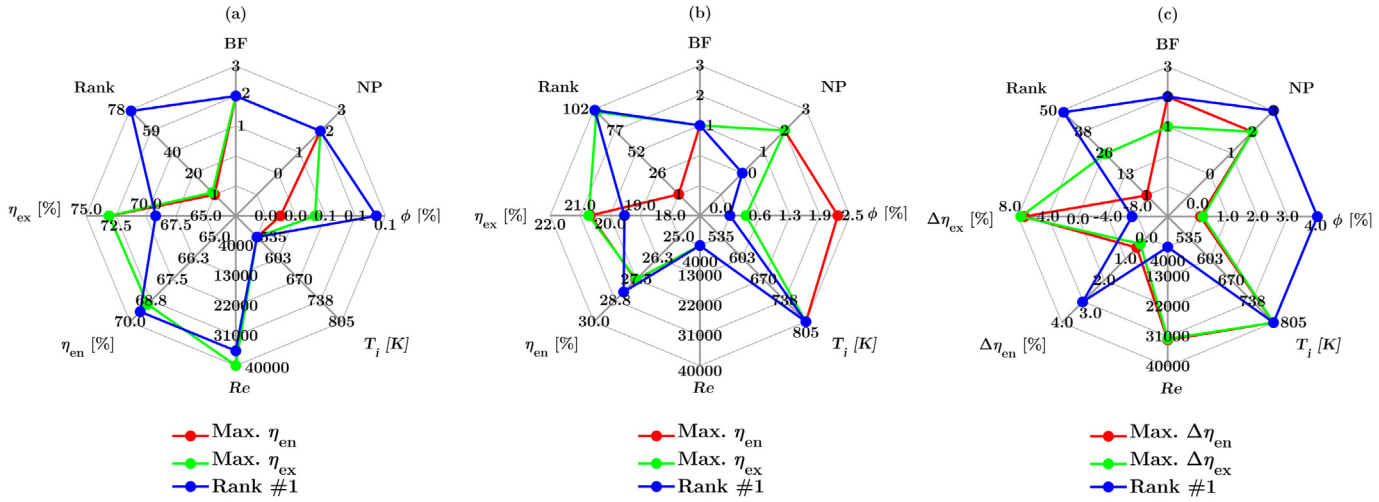


Fig. 13. Selected solutions from the global Pareto front for (a) entire studied domain of operating conditions with efficiencies as the objective functions, (b) domain constrained at T_i = 805 K and Re = 4000 with efficiencies as the objective functions, and (c) the entire studied domain of operating conditions with percentage enhancements in the two efficiencies as the objective functions.

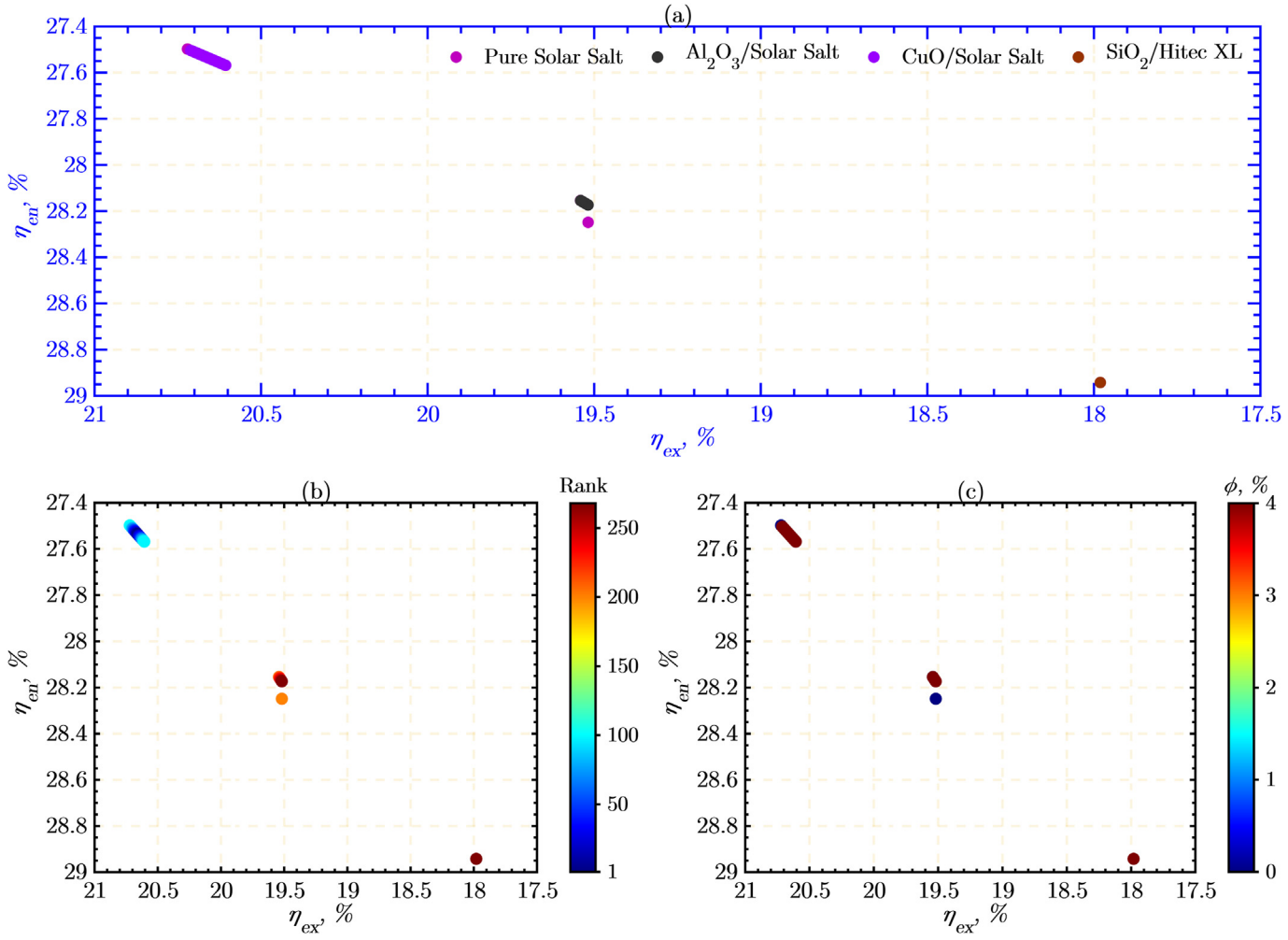


Fig. 14. Results of the optimization process for fixed operation at T_i = 805 K and Re = 4000. (a) Global Pareto solutions, (b) TOPSIS-based ranks, and (c) mapped global Pareto solutions in terms of nanoparticle concentration.

the salt itself. Hence, using an alternative salt might not be an easily feasible solution. For this, Fig. 15 focuses on the relative enhancement ($\Delta\eta_{en}$ and $\Delta\eta_{ex}$) in the PTC performance operating with nanofluids, compared to their corresponding base fluid.

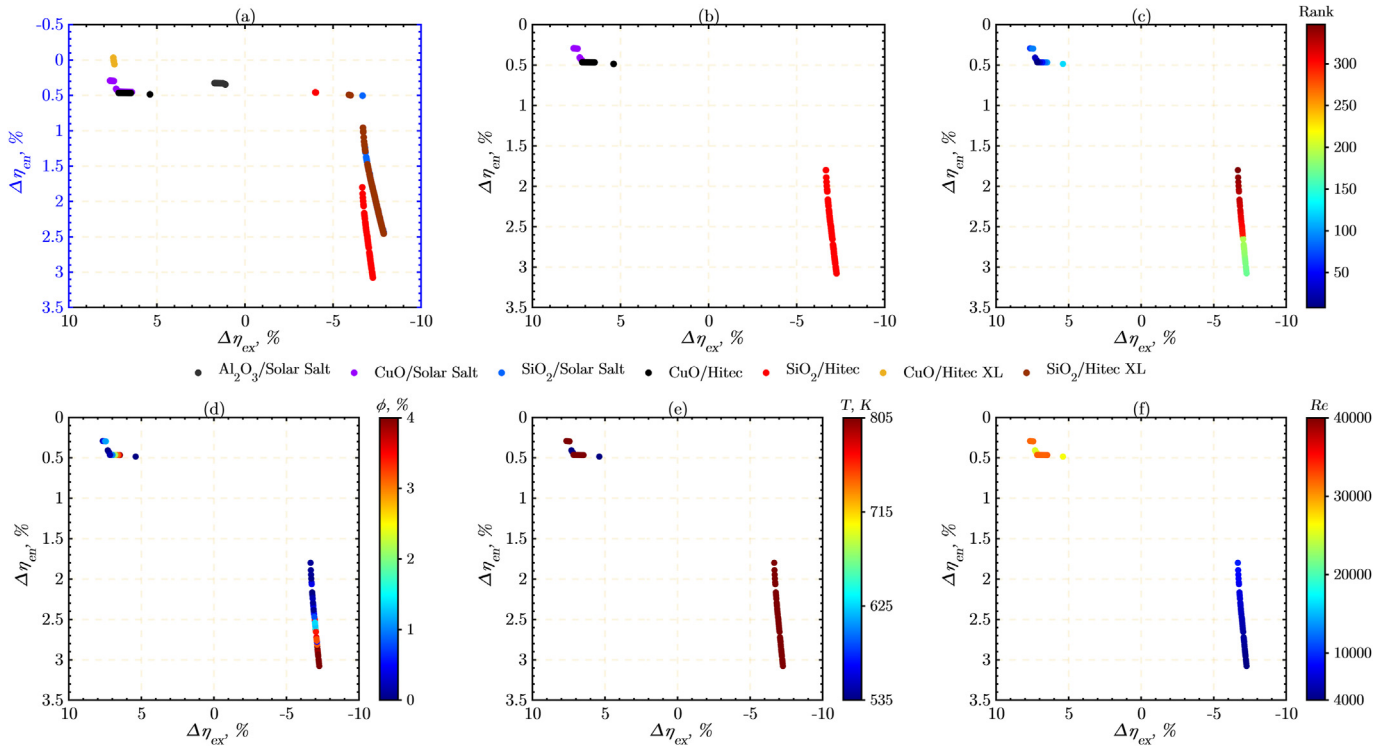


Fig. 15. Results of the multi-criteria optimization of relative performance. (a) Individual Pareto fronts, (b) global Pareto front, (c) TOPSIS-based ranks of the global Pareto solutions, (d)–(f) global Pareto solutions mapped based on the volumetric concentration, fluid inlet temperature, and Reynolds number, respectively.

Comparing Figs. 15(b) and Figure 12(b), some key differences can be spotted. First, the task of choosing the best nanofluid is more complicated. This is because no single nanofluid dominates the others. Fig. 15(b) shows that the final global Pareto front consists of three different nanofluids, namely: CuO/Solar Salt, Al_2O_3/Hitec, and CuO/Hitech, whereas in Fig. 12(b), only CuO/Hitech made an appearance. Next, the operating temperatures in Fig. 15(b) are much higher. This again emphasizes that the aforementioned related conclusions in Fig. 14. Also, the same can be said for *Re* in Fig. 15(c). Finally, Fig. 15(d) shows a much higher and more versatile range of nanoparticle concentration. Based on equal weights, the optimum final solution achieved percentage enhancements in energy and exergy efficiencies of 0.465 and 7.165%, respectively. This corresponds to CuO-Hitec nanofluids operating at T_i, *Re*, and ϕ of 805 K, 32025.4, and 0.0918%, respectively. Fig. 13(a)–(c) summarizes the above-mentioned differences between $\Delta\eta$- and η-based optimizations for the three key solutions from both the single and multi-objective optimization perspectives.

8. Conclusions

This study was concerned with mapping and optimizing the design and operating conditions of molten salt-based nanofluids using a unique integrated MCRT-FVM-SVR-PSO-TOPSIS framework. The impact of utilizing different nanoparticles in different molten salts under broad ranges of thermal and hydraulic conditions was examined in terms of thermal, hydraulic, energetic, and exergetic performances of parabolic trough concentrators. The key findings of the study can be summarized as follows.

1 The type of molten salt is more influential than the type of nanoparticles. Before deciding to use MSNFs, a first decision should be made on whether another salt type can be considered based on other environmental and economic aspects. In this context, Hitec molten salt was found more effective than Solar Salt and Hitec XL and even more than their MSNFs.
2 The PTC's energy efficiency was found to increase by increasing the flow rate and nanoparticle concentration and decreasing the operating temperature. On the other hand, PTC's exergy efficiency was found to increase by increasing the flow rate and decreasing the operating temperature and nanoparticle concentration.
3 Among the different simulated cases using the MCRT-FVM, the maximum achieved energy efficiency was ~69.1%, which was achieved when operating the PTC using SiO_2-Hitec nanofluid (volumetric concentration of 1.0%) at a Reynolds number of 40×10^4 and inlet temperature of 535 K. Meanwhile, the maximum achieved exergy efficiency was 70.477%, obtained when operating using pure Hitec at a Reynolds number of 40×10^4 and temperature of 535 K.
4 Solar Salt showed a wider range of operating conditions and nanoparticle concentrations where the use of MSNF is effective, with enhancements up to ~17%. Narrower ranges were noticed for Hitec, and a very limited range was noticed for Hitec XL. In terms of relative exergetic performance, only Hitec-based nanofluids showed the potential of enhancing the exergy efficiency up to ~42.0%.
5 Based on equally weighted criteria of energetic and exergetic performances, the optimum energy and exergy efficiencies of the multi-objective optimization are ~73.1 and 69.0%, respectively. This corresponds to operating temperature, Reynolds number, and nanoparticle concentration of 535 K, 39912.98, and 0.019%, respectively. The optimum final solution of the relative enhancement optimization problem showed enhancements in energy and exergy efficiencies of 0.465 and 7.166%, respectively, which corresponds to CuO-Hitec nanofluids operating at T_i, *Re*, and ϕ of 805 K, 32025.4, and 0.092%, respectively.

Based on the presented results, the worthiness of utilizing MSNFs to boost the PTC performance depends on many factors such as the working conditions and the type of the base salt. This study not only represents a viable tool to help the decision-makers make the most appropriate decision based on their needs but also demonstrates tailorable procedures and methodology that can be adjusted for other domains aside from the ones focused on in the current study. The proposed framework can be adopted for selecting the nanofluid types and the exact operational conditions for maximizing either the system performance or the performance enhancement over plain molten salts. The framework can also be utilized with conventional base fluid or extended to optimize other performance parameters. As future work, it is recommended to extend the proposed framework as a larger investigation of the technical (through whole year simulations using simplified models), economic, and environmental aspects of the system.

Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

CRediT authorship contribution statement

Amr Kaood: Software, Formal analysis, Investigation, Data curation, Visualization, Data curation, Software. **Mohamed Abubakr:** Methodology, Software, Validation, Visualization, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Otabeh Al-Oran:** Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Muhammed A. Hassan:** Conceptualization, Methodology, Validation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

All results of the simulations using the MCRT-FVM model are provided in the supplementary material attached to the online version of the article.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2021.06.049.

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